

DBMS LAB ASSIGNMENT 4, 5 and 6

```
CREATE TABLE Students (  
    Student_ID INT,  
    Student_Name VARCHAR(50),  
    Class VARCHAR(20),  
    Balance DECIMAL(10,2),  
    Admission_Date DATE,  
    Marks INT,  
    Fee_Amount DECIMAL(10,2),  
    Date_of_Birth DATE  
);  
  
INSERT INTO students (Student_ID, Student_Name, Class, Balance, Admission_Date, Marks, Fee_Amount, Date_of_Birth)  
VALUES  
(301, 'Ruthu Gowda', '10A', 68000, '2023-06-08', 97, 55000, '2008-03-15'),  
(302, 'Mrudula Bhat', '10B', 92000, '2023-06-12', 90, 60000, '2008-07-22'),  
(303, 'Dhrithi Sharam', '10A', 47000, '2023-06-15', 87, 48000, '2009-01-10'),  
(304, 'Vashanvi Guptha', '10C', 85000, '2023-06-18', 96, 52000, '2008-10-05'),  
(305, 'Simtha Malhotra', '10B', 59000, '2023-06-20', 91, 45000, '2009-02-28');  
  
SELECT * FROM students;  
  
SELECT Student_Name,  
UPPER(Student_Name) AS Uppercase_Name  
FROM Students;  
  
SELECT Student_Name,  
LOWER(Student_Name) AS Lowercase_Name  
FROM Students;  
  
SELECT Student_Name,  
LENGTH(Student_Name) AS Name_Length  
FROM Students;  
  
SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details  
FROM Students;  
  
SELECT Student_Name,  
SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters  
FROM Students;
```

```
SELECT Student_Name,  
Marks,  
ROUND(Marks * 1.05) AS Bonus_Marks  
FROM Students;
```

```
SELECT Student_Name,  
Marks / 10.0 AS Marks_Value,
```

```
CEIL(Marks / 10.0) AS Rounded_Up  
FROM Students;
```

```
SELECT Student_Name,  
Marks / 10.0 AS Marks_Value,  
FLOOR(Marks / 10.0) AS Rounded_Down  
  
FROM Students;
```

```
SELECT Student_Name,  
Marks,  
MOD(Marks, 10) AS Remainder  
FROM Students;
```

```
SELECT Student_Name,  
Marks,  
ABS(Marks - 80) AS Difference_From_80  
FROM Students;
```

```
SELECT Student_Name,  
Date_of_Birth,  
YEAR(Date_of_Birth) AS Birth_Year  
FROM Students;
```

```
SELECT Student_Name,  
Date_of_Birth,  
MONTH(Date_of_Birth) AS Birth_Month  
FROM Students;
```

```
SELECT Student_Name,  
Date_of_Birth,  
MONTH(Date_of_Birth) AS Birth_Month  
FROM Students;
```

```
SELECT Student_Name,  
Date_of_Birth,  
DAY(Date_of_Birth) AS Birth_Day  
FROM Students;
```

```
SELECT Student_Name,  
Date_of_Birth,  
TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age  
FROM Students;
```

```
SELECT CURDATE() AS Today;
```

```
SELECT COUNT(*) AS Total_Students  
FROM Students;
```

```
SELECT SUM(Marks) AS Total_Marks  
FROM Students;
```

```
SELECT AVG(Marks) AS Average_Marks  
FROM Students;
```

```
SELECT MAX(Marks) AS Highest_Marks  
FROM Students;
```

```
SELECT MIN(Marks) AS Lowest_Marks  
FROM Students;
```

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_Name	Uppercase_Name
Ruthu Gowda	RUTHU GOWDA
Mrudula Bhat	MRUDULA BHAT
Dhrithi Sharam	DHRITHI SHARAM
Vashanvi Guptha	VASHANVI GUPTHA
simtha Malhotra	SIMTHA MALHOTRA

Student_Name	Lowercase_Name
Ruthu Gowda	ruthu gowda
Mrudula Bhat	mrudula bhat
Dhrithi Sharam	dhrithi sharam
Vashanvi Guptha	vashanvi guptha
simtha Malhotra	simtha malhotra

Student_Name	Name_Length
Ruthu Gowda	11
Mrudula Bhat	12
Dhrithi Sharam	14
Vashanvi Guptha	15
simtha Malhotra	15

Student_Details
Ruthu Gowda - Class 10A
Mrudula Bhat - Class 10B
Dhrithi Sharam - Class 10A
Vashanvi Guptha - Class 10C
simtha Malhotra - Class 10B

Student_Name	First_Five_Characters
Ruthu Gowda	Ruthu
Mrudula Bhat	Mrudu
Dhrithi Sharam	Dhrit
Vashanvi Guptha	Vasha
simtha Malhotra	simth

Student_Name	Marks	Bonus_Marks
Ruthu Gowda	97	102
Mrudula Bhat	90	95
Dhrithi Sharam	87	91
Vashanvi Guptha	96	101
simtha Malhotra	91	96

Student_Name	Marks_Value	Rounded_Up
Ruthu Gowda	9.7000	10
Mrudula Bhat	9.0000	9
Dhrithi Sharam	8.7000	9
Vashanvi Guptha	9.6000	10
simtha Malhotra	9.1000	10

Student_Name	Marks_Value	Rounded_Down
Ruthu Gowda	9.7000	9
Mrudula Bhat	9.0000	9
Dhrithi Sharam	8.7000	8
Vashanvi Guptha	9.6000	9
simtha Malhotra	9.1000	9

Student_Name	Marks	Remainder
Ruthu Gowda	97	7
Mrudula Bhat	90	0
Dhrithi Sharam	87	7
Vashanvi Guptha	96	6
simtha Malhotra	91	1

Student_Name	Marks	Difference_From_80
Ruthu Gowda	97	17
Mrudula Bhat	90	10
Dhrithi Sharam	87	7
Vashanvi Guptha	96	16
simtha Malhotra	91	11

Student_Name	Date_of_Birth	Birth_Year
Ruthu Gowda	2008-03-15	2008
Mrudula Bhat	2008-07-22	2008
Dhrithi Sharam	2009-01-10	2009
Vashanvi Guptha	2008-10-05	2008
simtha Malhotra	2009-02-28	2009

Student_Name	Date_of_Birth	Birth_Month
Ruthu Gowda	2008-03-15	3
Mrudula Bhat	2008-07-22	7
Dhrithi Sharam	2009-01-10	1
Vashanvi Guptha	2008-10-05	10
simtha Malhotra	2009-02-28	2

Student_Name	Date_of_Birth	Birth_Day
Ruthu Gowda	2008-03-15	15
Mrudula Bhat	2008-07-22	22
Dhrithi Sharam	2009-01-10	10
Vashanvi Guptha	2008-10-05	5
simtha Malhotra	2009-02-28	28

Student_Name	Date_of_Birth	Age
Ruthu Gowda	2008-03-15	18
Mrudula Bhat	2008-07-22	18
Dhrithi Sharam	2009-01-10	17
Vashanvi Guptha	2008-10-05	17
simtha Malhotra	2009-02-28	17

+-----+	
Today	
+-----+	
2026-10-02	
+-----+	
+-----+	
Total_Students	
+-----+	
5	
+-----+	
+-----+	
Total_Marks	
+-----+	
461	
+-----+	
+-----+	
Average_Marks	
+-----+	
92.2000	
+-----+	
+-----+	
Highest_Marks	
+-----+	
97	
+-----+	
+-----+	
Lowest_Marks	
+-----+	
87	
+-----+	

```
SELECT Student_Name,
       CAST(Marks AS DECIMAL(10,2)) AS Converted_Marks
FROM Students;
```

```
SELECT Student_Name,
       CAST(Balance AS CHAR) AS Balance_As_Text
FROM Students;
```

```
SELECT Student_Name,
       Fee_Amount + 5000 AS Increased_Fee
FROM Students;
```

```
SELECT Student_Name,
       Fee_Amount + 5000 AS Increased_Fee
FROM Students;
```

```
SELECT Student_Name,
       Fee_Amount - 5000 AS Reduced_Fee
FROM Students;
```

```
SELECT Student_Name,
       Fee_Amount * 2 AS Double_Fee
FROM Students;
```

```
SELECT Student_Name,
       Fee_Amount / 2 AS Half_Fee
FROM Students;
```

```
SELECT *
FROM Students
WHERE Marks > 90;
```

```
SELECT *
FROM Students
WHERE Marks < 90;
```

```
SELECT *
FROM Students
WHERE Class = '10A';
```

```
SELECT *
FROM Students
```

STDIN

Input for the program (Optional)

Output:

Student_Name	Converted_Marks
Ruthu Gowda	97.00
Mrudula Bhat	90.00
Dhrithi Sharam	87.00
Vashanvi Guptha	96.00
simtha Malhotra	91.00

Student_Name	Balance_As_Text
Ruthu Gowda	68000.00
Mrudula Bhat	92000.00
Dhrithi Sharam	47000.00
Vashanvi Guptha	85000.00
simtha Malhotra	59000.00

Student_Name	Increased_Fee
Ruthu Gowda	60000.00
Mrudula Bhat	65000.00
Dhrithi Sharam	53000.00
Vashanvi Guptha	57000.00
simtha Malhotra	50000.00

Student_Name	Increased_Fee
Ruthu Gowda	60000.00

```

SELECT *
FROM Students
WHERE Marks <> 90;

SELECT *
FROM Students
WHERE Marks >= 90;

SELECT *
FROM Students
WHERE Marks <= 90;

SELECT *
FROM Students
WHERE Marks > 90 AND Balance > 60000;

SELECT *
FROM Students
WHERE Marks > 95 OR Class = '10B';

SELECT *
FROM Students
WHERE NOT Class = '10A';

SELECT *
FROM Students
WHERE NOT Class = '10A';

SELECT *
FROM Students
WHERE Marks BETWEEN 90 AND 97;

SELECT *
FROM Students
WHERE Class IN ('10A', '10B');

SELECT *
FROM Students
WHERE Class NOT IN ('10A', '10B');

```

```

FROM Students
WHERE Class IN ('10A', '10B');

SELECT *
FROM Students
WHERE Class NOT IN ('10A', '10B');

SELECT *
FROM Students
WHERE Student_Name LIKE 'R%';

SELECT *
FROM Students
WHERE Student_Name LIKE '%a';

SELECT *
FROM Students
WHERE Balance IS NULL;

SELECT *
FROM Students
WHERE Balance IS NOT NULL;

SELECT Student_Name, Class
FROM Students
WHERE Class = '10A'

UNION

SELECT Student_Name, Class
FROM Students
WHERE Class = '10B';

SELECT Student_Name, Class
FROM Students
WHERE Class = '10A'

UNION ALL

SELECT Student_Name, Class
FROM Students
WHERE Class = '10B';

```

STDIN

Input for the program (Optional)

Output:

Student_Name	Increased_Fee
Ruthu Gowda	60000.00
Mrudula Bhat	65000.00
Dhrithi Sharam	53000.00
Vashanvi Guptha	57000.00
simtha Malhotra	50000.00

Student_Name	Reduced_Fee
Ruthu Gowda	50000.00
Mrudula Bhat	55000.00
Dhrithi Sharam	43000.00
Vashanvi Guptha	47000.00
simtha Malhotra	40000.00

Student_Name	Double_Fee
Ruthu Gowda	110000.00
Mrudula Bhat	120000.00
Dhrithi Sharam	96000.00
Vashanvi Guptha	104000.00
simtha Malhotra	90000.00

STDIN

Input for the program (Optional)

Output:

Student_Name	Half_Fee
Ruthu Gowda	27500.000000
Mrudula Bhat	30000.000000
Dhrithi Sharam	24000.000000
Vashanvi Guptha	26000.000000
simtha Malhotra	22500.000000

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10

Output:

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Output:

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15

Output:

301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28
Student_Name	Class						
Ruthu Gowda	10A						
Dhrithi Sharam	10A						
Mrudula Bhat	10B						
simtha Malhotra	10B						
Student_Name	Class						
Ruthu Gowda	10A						
Dhrithi Sharam	10A						
Mrudula Bhat	10B						
simtha Malhotra	10B						